

Offshore DL Dissertation Results Snapshot – May 2026

Generated from the May 2026 claim ledger and final table pack

2026-05-28

Scope and claim controls

This snapshot summarizes current dissertation-facing benchmark evidence from the final table pack sourced at revision 44402db9fc6e and refreshed on repository base ca6a042cfd82. Claims are controlled by `reports/dissertation_claim_ledger_2026-05.md`; tables are transcribed from `reports/dissertation_final_tables_2026-05.md`. Stage 1 3W and Stage 2 3W are separate benchmark families. CDF trained reconstruction and foundation forecast metrics are not pooled. Forecasting conclusions are metric-specific.

T1 — 3W Stage 1 validated HPO, standard 720-window classification

Primary metric: held-out macro-F1. This is the standard apples-to-apples 720-window classification benchmark.

Rank	Model	Macro-F1	Accuracy	Trials
1	Random Forest	0.968972	0.970228	30
2	DeepONet	0.962579	0.968734	30
3	MambaSL	0.962185	0.967313	30
4	LSTM	0.960046	0.965314	30
5	FKMAD	0.956388	0.964061	30
6	ConvTimeNet	0.954953	0.959894	30
7	PatchTST	0.953556	0.958738	30

T2 — 3W Stage 2 separate follow-up variants

Stage 2 changes feature/window assumptions; report it separately from Stage 1.

Variant	Macro-F1	Accuracy	Validity note
window360_rf	0.987797	0.991537	Valid follow-up; different window length from Stage 1.
window1440_rf	0.977011	0.989084	Valid follow-up; different window length from Stage 1.
wavelet_rf	0.964309	0.966301	Valid feature variant.
multiscale_rf	0.964184	0.966205	Valid feature variant.
physics_rf	0.964070	0.966060	Valid feature variant.
wavelet_deeponet	0.961252	0.967867	Valid DeepONet variant.
convtran	0.955762	0.964398	Valid completed baseline.
multiscale_deeponet	0.954235	0.964880	Valid DeepONet variant.
physics_deeponet	0.952286	0.961652	Valid DeepONet variant.

T3 — Ganymede post-fix multi-well forecasting

Arithmetic means across h7/h14/h30/h90 multi-well rows. Lower is better for MAE/RMSE/MASE; higher is better for R^2 diagnostics.

Model	MAE	RMSE	MASE	R^2	R^2_{prod}
TiRex	0.3620	1.2481	0.2072	0.3537	-0.1499
TimesFM	0.3967	1.2640	0.2297	0.3356	-0.1293
Chronos-2	0.5373	1.4426	0.3215	0.1094	-0.3438
LSTM	0.5457	1.3519	0.0228	0.2453	-0.6491
TCN	0.5680	1.3140	0.0234	0.2860	-0.4109
DeepONet	0.6794	1.3748	0.0301	0.2219	-0.4026
PatchTST	1.0774	2.1175	0.0474	-0.8660	-1.2016

Metric-specific interpretation. TiRex leads Ganymede by MAE/RMSE and R^2 diagnostics, while LSTM/TCN lead grouped MASE-style scaled diagnostics. Do not collapse these into one universal forecasting winner.

T4 — Cross-dataset forecasting Borda diagnostics

Lower Borda score is better. These are ranking diagnostics, not raw-scale metric averages.

Metric	Rank 1	Rank 2	Rank 3
MAE	TiRex 1.660	Chronos-2 2.212	TimesFM 3.230
R^2_{prod}	TiRex 2.588	Chronos-2 3.077	TimesFM 3.442
MASE	PatchTST 2.619	LSTM 3.171	TCN 3.427

T5 — Forecasting sparse h90 per-well exclusions

These missing h90 per-well rows are data-coverage exclusions, not active HPC failures.

Dataset	Horizon	Mode	Missing scenarios	Missing rows
inner_mongolia	90	per_well	57-14X, 57-15X	14
spe_berg	90	per_well	well_11, well_2	14
volve	90	per_well	NO_15_9-F-5_AH	7

T6 — CDF trained reconstruction metrics

Lower is better. Compare only within trained reconstruction semantics.

Model	error_mean	error_p50	error_p95	error_p99
LSTM	0.005878	0.005289	0.007909	0.015403
PatchTST	0.081621	0.071603	0.150840	0.271780
DeepONet	0.230735	0.223694	0.414683	0.425859

T7 — CDF foundation forecast metrics

Lower is better. Compare only within one-step foundation forecast semantics.

Model	forecast_error_mean	forecast_error_p50	forecast_error_p95	forecast_error_p99
Chronos-2	0.243968	0.226866	0.457304	0.565650
TiRex	0.268548	0.246441	0.501056	0.695557
TimesFM	0.295999	0.283747	0.496752	0.589213

Usage notes

T1 and T2 answer different classification questions. T4 is diagnostic and metric-specific. T6 and T7 use different anomaly semantics and must never be pooled into one CDF ranking. Historical `results/pre_fix/` outputs remain audit lineage only.